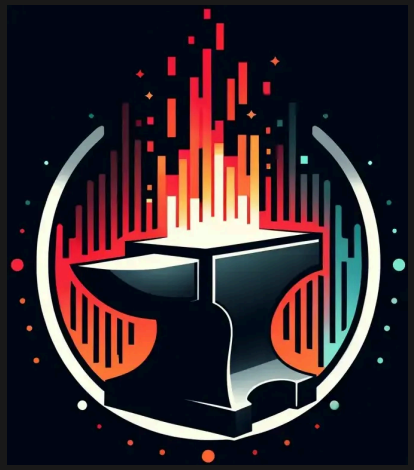
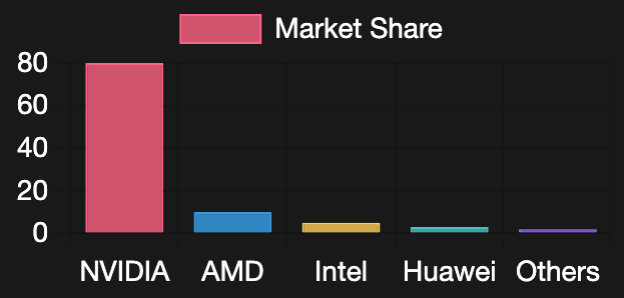


SMITH

Unleashing Neural Network Performance Across All Hardware

THE CHALLENGE

- Diverse hardware ecosystems with limited interoperability
- Dominance of specific APIs like CUDA, restricting flexibility
- Customers often forced to rely on NVIDIA GPUs



CURRENT STATE OF THE ART

- TensorFlow and PyTorch are the leading frameworks in the AI landscape
- Depend on manually optimized kernels, leading to limited flexibility
- High performance often confined to specific hardware platforms, such as NVIDIA GPUs
- Lack of interoperability between different hardware ecosystems

	Nvidia	Amd	Intel	Huawei	...	Apple
Kernel 1	✓	✓		✓		
Kernel 2	✓	✓		✓		✓
Kernel 3	✓		✓	✓		✓
Kernel 4	✓		✓			✓
...						
Kernel N		✓	✓	✓		✓

INTRODUCING SMITH

- Compiled language for high performance on CPUs and GPUs
- Advanced static type system tracking element types and dimensions
- Shape checking at compile time
- Source-to-source automatic differentiation at compile time
- Immutability by default for safe parallelism without locks
- Generalized Einstein summation notation for linear algebra

OPEN SOURCE STRATEGY

- Smith is open-source, licensed under the Mozilla Public License 2.0 (MPL 2.0)
- Facilitates commercial use, allowing seamless integration into products
- Encourages sharing improvements with the community
- Enriches the ecosystem and fosters greater innovation

ROADMAP AND MILESTONES

- Milestone 1: WebAssembly for CPU support
- Milestone 2: Compile-time automatic differentiation
- Milestone 3: WebGPU for GPU support

FUNDING OBJECTIVES

- Secure **$2.5 million upfront** to initiate full-time development of Smith
- Allocate funding primarily for development costs
- Additional funding based on achievement of key milestones
- Ensure continuous progress and innovation

JOIN SMITH

- Be part of the next revolution in neural networks
- Drive innovation in AI across diverse hardware platforms
- Expand your market reach and influence in the AI community

Contact us to learn more and become a partner in pioneering the future of AI.

KEY TECHNOLOGY STACK

- Tree-Sitter can parse the Smith language and provide syntax highlighting across all IDEs.
- LLVM allows me to generate code for various architectures including X86, X86-64, PowerPC, PowerPC-64, ARM, Thumb, SPARC, Alpha, CellSPU, MIPS, MSP430, SystemZ, WebAssembly, and XCore.
- wgpu lets me generate graphics using Vulkan, Metal, DirectX 12, OpenGL ES, WebGPU, and WebGL2.
- OpenXLA supports generation for AWS Tranium & Inferentia, AMD GPU, Arm CPUs, Cerebras WSE, Google TPU, Graphcore IPU, NVIDIA GPU, and x86 CPU.
- Language Server Protocol (LSP) provides features in every IDE such as go to definition, go to references, autocomplete, and documentation.
- Debug Adapter Protocol (DAP) offers debugging and running your program and tests in every IDE.